Amazon Web Services: Route 53

Topics

- What is DNS?
- IPv4 vs. IPv6
- Top-level Domains
- Domain Registrars
- Common DNS Record Types
- What is a TTL
- Alias Records
- Routing Policies
- Demo

What is DNS?

- Domain Name System (DNS) is the phonebook of the Internet
- Is used to convert huma-friendly domain names (e.g.. www.smu.edu into an Internet Protocol (IP) address (e.g., http://129.119.70.166)
- Two forms of IP addresses: IPv4 and IPv6

https://www.cloudflare.com/learning/dns/what-is-dns/

IPv4 vs. IPv6

IPv4 Addresses Are Running Out

IPv4

- The IPv4 space is a 32-bit field
- Has over 4 billion different addresses (4,294,967,296 – to be precise)

VS

IPv6

- IPv6 was created to solve this depletion issue
- Has an address space of 128 bits
- There are 340 undecillion addresses (a number equal to 1 followed by 36 zeros)

IPv4 vs. IPv6

IPv4 Addresses Are Running Out

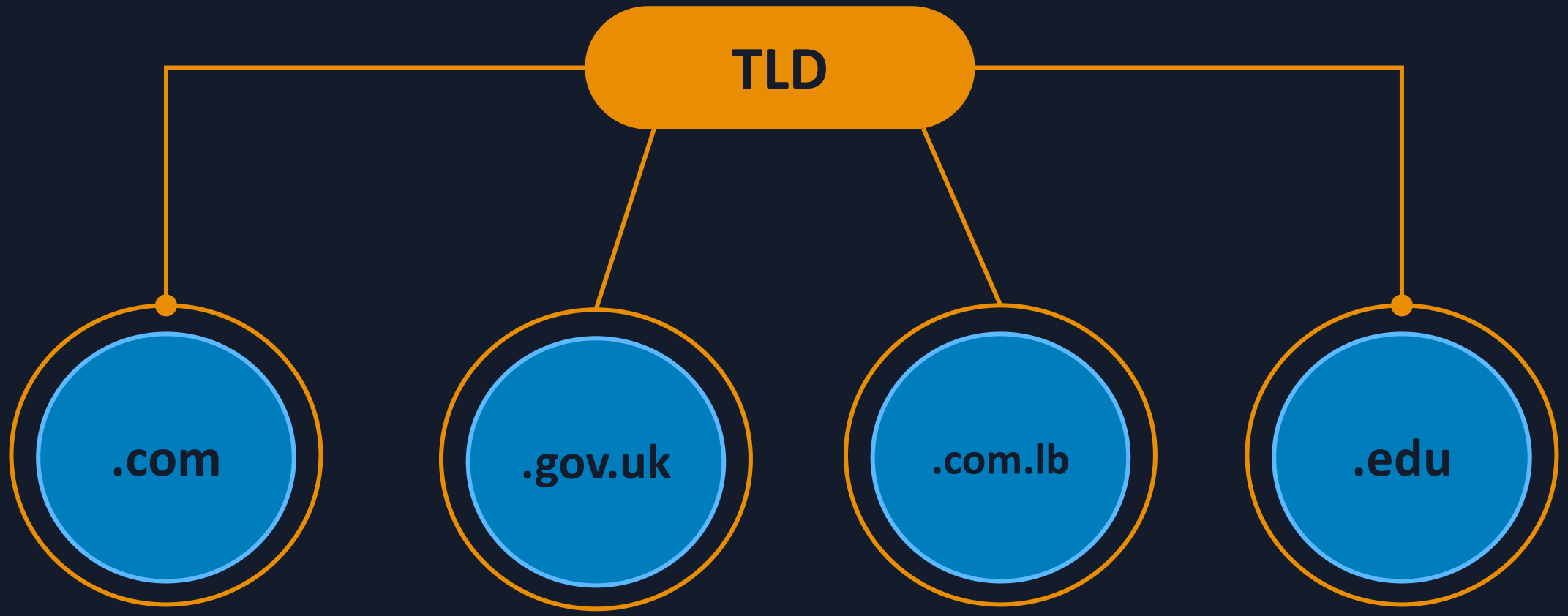
IPv4

IPv6

VS

Top-level Domain

- google.com, smu.edu, bbc.co.uk, …
- Controlled by the Internet Assigned Numbers Authority (IANA)
- https://www.iana.org/domains/root/db



Domain Registrars

- A way to organize domain names to ensure uniqueness
- A registrar is an authority that can assign domain names under one or more top-level domains
- Domains are registered with InterNIC, a service of ICANN, which enforces domain name uniqueness
- Each domain becomes registered in the WHOIS database

Examples of Domain Registrars (Activity)

5 Popular Domain Registrars

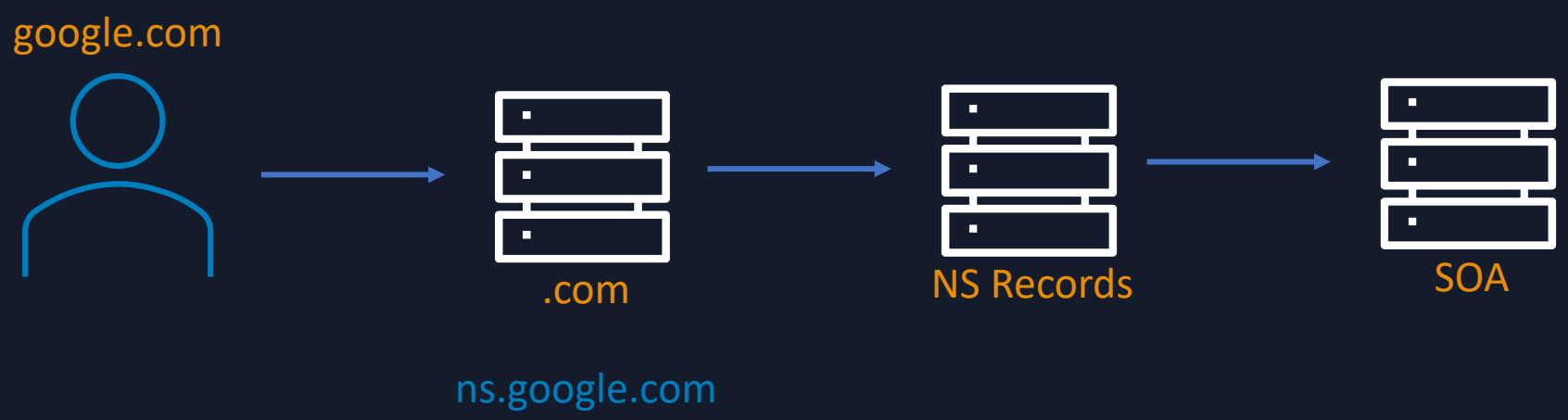
1. domain.com
2. GoDaddy
3. AWS
4. SquareSpace
5. Namecheap

Common DNS Record Types: SOA

- Start of Authority = SOA
- Stores admin information about a domain:
  - Name of the server that supplied the data for the zone
  - Administrator of the zone (email address)
  - Current version of the data files
  - Default number of seconds for the time to live files on the resource records

Name Server Records = NS

- Used by top level domain servers to direct traffic to content DNS servers that contain the authoritative DNS records



What is an A Record?

- Is the fundamental type of DNS record
- Used by a computer to translate the name of the domain to an IP address
- For example: https://www.smu.edu points to https://129.119.70.166

What is TTL?

- The length that a DNS record is cached on either the resolving server or the user's own local PC is equal to the value of the time to live (TTL) in seconds

- The lower the time to live, the faster the changes to DNS records take to propagate throughout the internet

What is a CNAME?

- A CNAME is a canonical name
- Can be used to resolve one domain to another

Alias Records

- Alias records are used to map resource record sets in your hosted zone to load balancers, CloudFront distributions, or S3 buckets that are configured as websites
- Alias records work like a CNAME record in that we can map one DNS name (www.example.com) to another target DNS name (elb1234.elb.amazonaws.com)

What is Route 53?

- Amazon's DNS service

- It allows you to register domain names, create hosted zones, and manage and create DNS records
- 8 Routing policies available with Route 53
- Simple, weighted, IP-based, latency-based, failover, geolocation, geoproximity, and multivalue answer

Amazon Route 53

https://docs.aws.amazon.com/Route53/latest/DeveloperGuide/routing-policy.html

Summary of Common DNS Record Types

- SOA Records
- CNAME Records
- NS Records
- A Records

Demo: Simple Routing Policy

- Can have one record with multiple IP addresses
- If you specify multiple values in a record, Route 53 returns all values to the user in a random order

Demo: Weighted Routing Policy

- Allows you to split your traffic based on different assigned weights
- Example: you can set 20% of your traffic to go to us-east-1 and 80% to go to us-west-1
- Can set health checks on individual records such as weightings

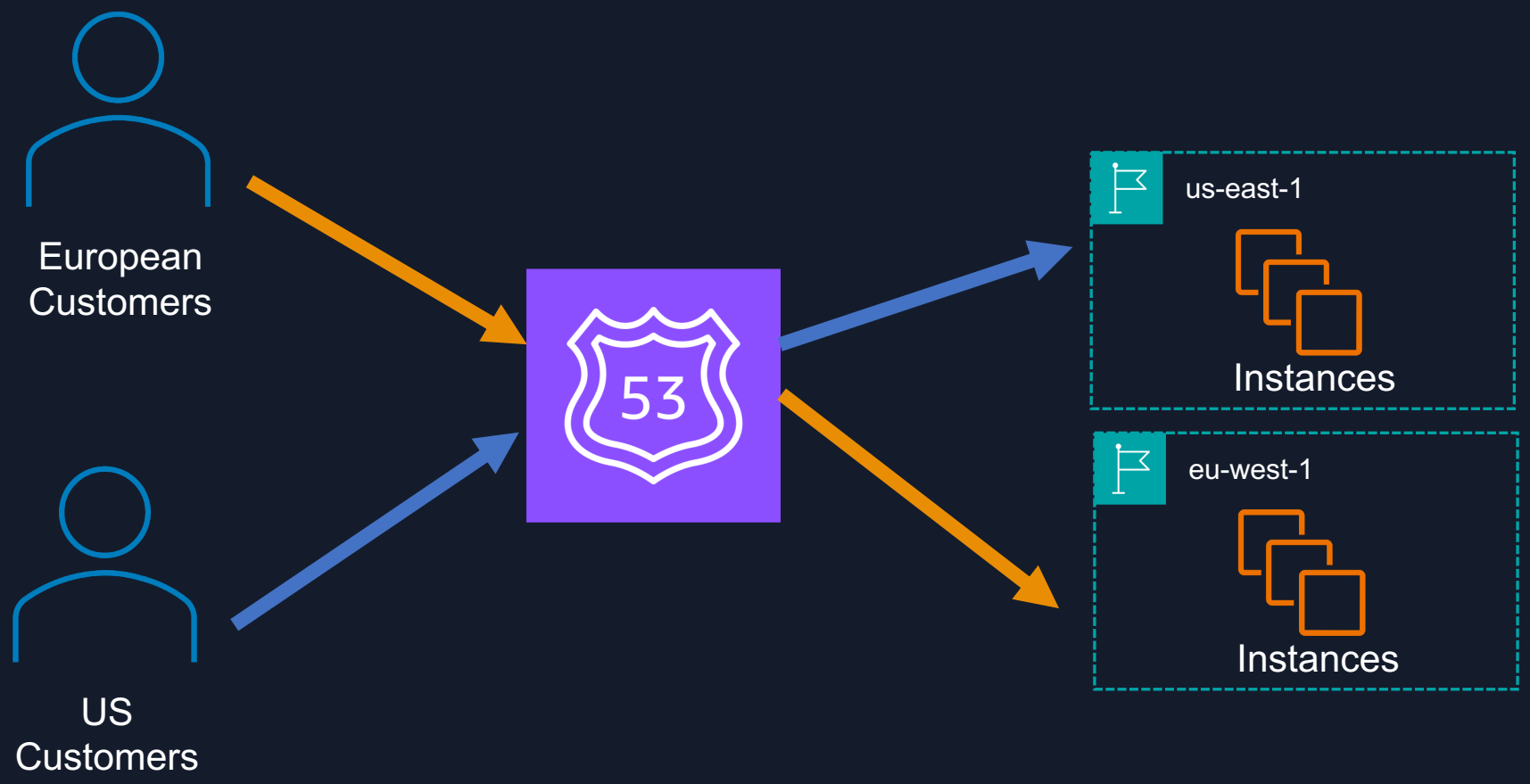
Demo: Failover Routing Policy

- Failover routing policies are used when you want to create an active/passive setup
- Route 53 will monitor the health of your primary site using a health check

Geolocation Routing Policy

- Allows you to choose where you traffic will be sent based on the geographic location of your users
  - i.e., the location from which the DNS queries originates
- Use cases: you might want all queries from Europe to be routed to a fleet of EC2 instances configures specifically for your European customers
- Localization: servers can have the local language of European customers and display prices in euros

Demo: Geolocation Routing Policy



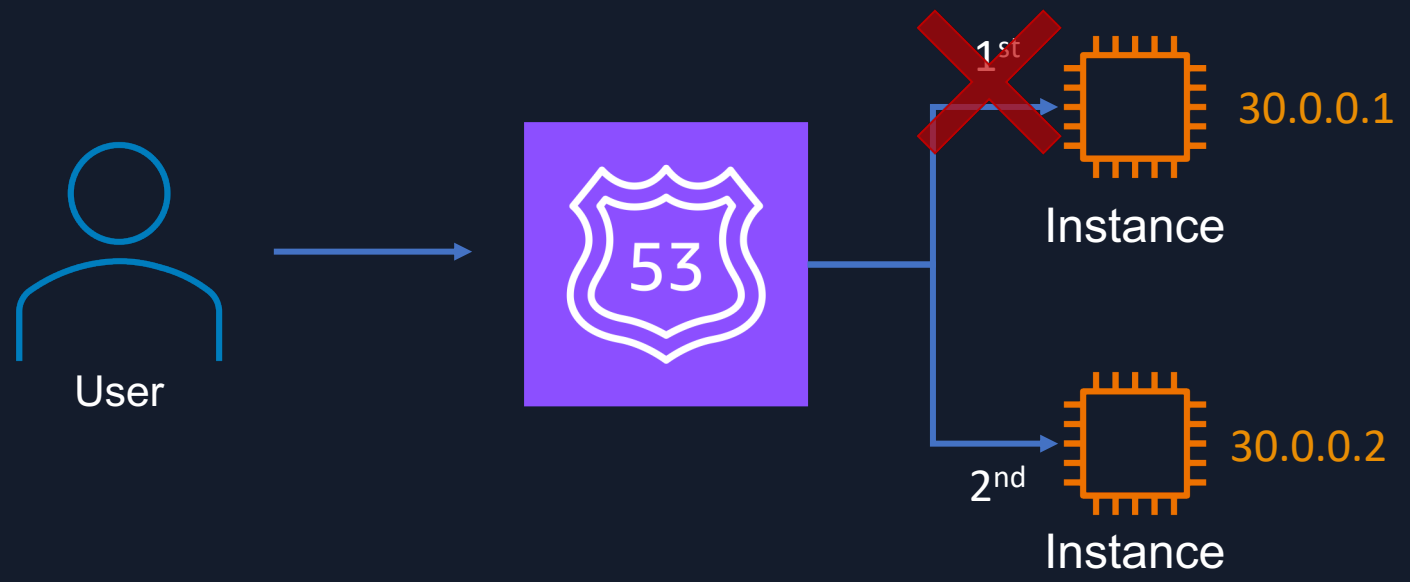
Latency Routing Policy

- Allows you to route your traffic based on the lowest network latency for your end user (i.e., which region will give them the fastest response time
- To use it, create a **latency resource record set** for the EC2 (or ELB) resource in each region that hosts your website
- When Route 53 receives a query for your site, it selects the latency resource record set for the region that gives the user the lowest latency
- Route 53 then responds with the value associated with that resource record set

Demo: Latency Routing Policy

Demo: Multivalue Answer Routing Policy

- Multivalue answer routing allows you configure Amazon Route 53 to return multiple values, such as IP addresses for your web servers, in response to DNS queries
- Like Simple Routing; however, it allows you to put health checks on each record set



Checkpoint

Which of the following statements about Route 53 failover policies is true?

1. With DNS failover, if a record set fails a security check, it will be removed from the Route 53 response until the owner adds it again.
2. With DNS failover, if a record set fails a health check, it will be removed from the Route 53 response until the owner adds it again.
3. With DNS failover, if a record set fails a security check, it will be removed from the Route 53 response until it passes the security check.
4. With DNS failover, if a record set fails a health check, it will be removed from the Route 53 response until it passes the health check.

Using Route 53, how would you direct "example.com" to a single IP address?

1. Use a weighted routing policy.
2. Use a simple routing policy.
3. Use a multi-value routing policy.
4. Use a failover routing policy.

Using Route 53, how would you direct "example.com" to 2 different IPs with different distribution levels of traffic?

1. Use a failover routing policy.
2. Use a simple routing policy.
3. Use a weighted routing policy.
4. Use a geolocation routing policy.

Using Route 53, how would you direct "example.com" traffic to 2 different IPs based on which one is most performant for a user?

1. Use a geolocation routing policy.
2. Use a latency routing policy.
3. Use a weighted routing policy.
4. Use a failover routing policy.

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